

$T\bar{T}$ deformation in large N limit

Junichi Haruna
(Kyoto Univ.)

2019/10/31 @ East Asia Joint Workshop

Based on work in progress with T Ishii, H Kawai, K Sakai, K Yoshida

Introduction

- ◇ $T\bar{T}$ deformation is an **irrelevant** deformation of 2D (classical) Lagrangian with the $T\bar{T}$ flow equation:

$$\frac{\partial \mathcal{L}^{(\alpha)}}{\partial \alpha} = \alpha \det(T_{\mu\nu}^{(\alpha)})$$

where $T_{\mu\nu}^{(\alpha)}$ is the energy-momentum tensor defined from $\mathcal{L}^{(\alpha)}$.

[Smirnov-Zamolodchikov, Cavaglià-Negro-Szécsényi-Tateo]

- ◇ This is studied in **various contexts**.
(Integrable QFT and integrability, coupling to topological gravity, Higher-dimensional generalization, Holography, Little string theory ...)

Introduction

- ◇ Classical aspects of the $T\bar{T}$ flow have been studied well. But **quantum** aspects of the $T\bar{T}$ flow are **not**.
 - ◇ It has been pointed out that by performing a field-dependent coordinate transformation, $T\bar{T}$ deformation can be undone while the base space is deformed with a classical metric perturbation.

[Cardy 2018, Dubovsky-Gorbenko-Mirbabayi 2017, Conti-Negro-Tateo 2018]
- ◇ If the $T\bar{T}$ flow can be understood at **quantum** level, a clue to **quantum gravity** may be captured.
- ◇ We discuss $T\bar{T}$ deformation at quantum level in large N limit, Especially we focus on **renormalizability** and **positivity** of state space.

What I will talk about

- ◆ We study infinitesimal $T\bar{T}$ deformation of **massive $O(N)$ vector model** in the large N limit.

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 + \frac{m^2}{2} \vec{\phi}^2 + \alpha \det(T_{\mu\nu})$$

- ◆ We find that

1. The deformation parameter(α) should be **renormalized**.
2. Logarithmic divergence does **not appear**.
3. **Positivity** of state space is **violated**.

If not, **non-local** counterterms are needed:

$$\mathcal{L}_{\text{c.t.}} \ni \alpha^2 T^{\mu\nu} \frac{\partial^2}{1 + \alpha \delta_z \partial^2} T_{\mu\nu}$$

$T\bar{T}$ in large N limit

- Consider the following Lagrangian:

$$\frac{\mathcal{L}}{N} = \frac{1}{2}(\partial_\mu \vec{\phi})^2 + \frac{m^2}{2}\vec{\phi}^2 + \alpha \det(T_{\mu\nu})$$
$$T_{\mu\nu} := \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} - \frac{\eta_{\mu\nu}}{2} \left((\partial_\rho \vec{\phi})^2 + m^2 \vec{\phi}^2 \right)$$

(For simplicity, we work in Euclidean spacetime.)

- This Lagrangian is a solution to the $T\bar{T}$ flow equation with “initial condition” of

$$\frac{\mathcal{L}^{(0)}}{N} = \frac{1}{2}(\partial_\mu \vec{\phi})^2 + \frac{m^2}{2}\vec{\phi}^2$$

up to $O(\alpha)$. [Bonelli-Doroud-Zhu]

- We calculate **the effective action** in $N \rightarrow \infty$ limit.

$T\bar{T}$ in large N limit

◆ Rewriting $\mathcal{L}$ with auxiliary fields $C, \tilde{C}^{\mu\nu}$:

C ... scalar field
 $\tilde{C}^{\mu\nu}$... traceless symmetric tensor field

$$\frac{\mathcal{L}}{N} = \frac{1}{2}(\partial_\mu \vec{\phi})^2 + \frac{m^2}{2}\vec{\phi}^2 - m^2 C \vec{\phi}^2 + \tilde{C}^{\mu\nu} \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} - \frac{1}{\alpha} C^2 + \frac{1}{2\alpha} (\tilde{C}^{\mu\nu})^2$$

(With E.O.M of $C, \tilde{C}^{\mu\nu}$, this reproduces the original $\mathcal{L}$)

◆ The effective action:

$$\begin{aligned} \frac{S_{\text{eff}}}{N} = & \frac{1}{2} \text{tr} \log(-\partial^2 + (1 - 2C)m^2 - 2\partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) \\ & + \int_x \frac{-1}{\alpha} C^2 + \frac{1}{2\alpha} (\tilde{C}^{\mu\nu})^2 \end{aligned}$$

$N \rightarrow \infty$ limit

- ◆ In $N \rightarrow \infty$ limit, the stationary point of S_{eff} dominates for C & $\tilde{C}^{\mu\nu}$ integral.

- ◆ Stationary condition (E.O.M or gap equation) for C :

$$-\frac{m^2}{8\pi^2} \int \frac{d^2k}{k^2 + (1 - 2\langle C \rangle)m^2} = \frac{1}{\alpha} \langle C \rangle$$

- Because L.H.S diverges logarithmically ($\sim \log \Lambda$),
 α should be renormalized.

- ◆ From translation and Lorentz invariance, $\langle \tilde{C}^{\mu\nu} \rangle = 0$.
This satisfies the stationary condition trivially.

1/N correction

◇ To consider $\frac{1}{N}$ (**sub-leading**) contributions,

we set $C = \langle C \rangle + \tilde{C}$ and expand S_{eff} up to **2nd order of $\tilde{C}, \tilde{C}^{\mu\nu}$** :

$$\frac{S_{\text{eff}}}{N} = \frac{1}{2} \int_{x,y} (\tilde{C}_{(x)} \tilde{C}_{(x)}^{\alpha\beta}) \begin{pmatrix} M & M_{\mu\nu} \\ M_{\alpha\beta} & M_{\mu\nu\alpha\beta} \end{pmatrix} \begin{pmatrix} \tilde{C}_{(y)} \\ \tilde{C}^{\mu\nu}_{(y)} \end{pmatrix}$$

$$M = -2m^4 G(x-y) G(y-x) - \frac{2}{\alpha} \delta(x-y)$$

$$M_{\mu\nu} = 2m^2 \frac{\partial}{\partial x^\mu} G(x-y) \frac{\partial}{\partial x^\nu} G(x-y)$$

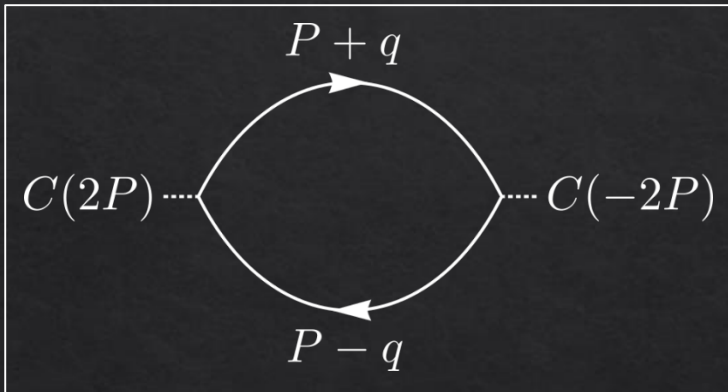
$$M_{\mu\nu\alpha\beta} = -2m^2 \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^\alpha} G(x-y) \frac{\partial}{\partial x^\nu} \frac{\partial}{\partial x^\beta} G(x-y) + \frac{\eta_{\mu\alpha} \eta_{\nu\beta}}{\alpha} \delta(x-y)$$

$$G(x) := (-\partial^2 + (1 - 2\langle C \rangle)m^2)^{-1}$$

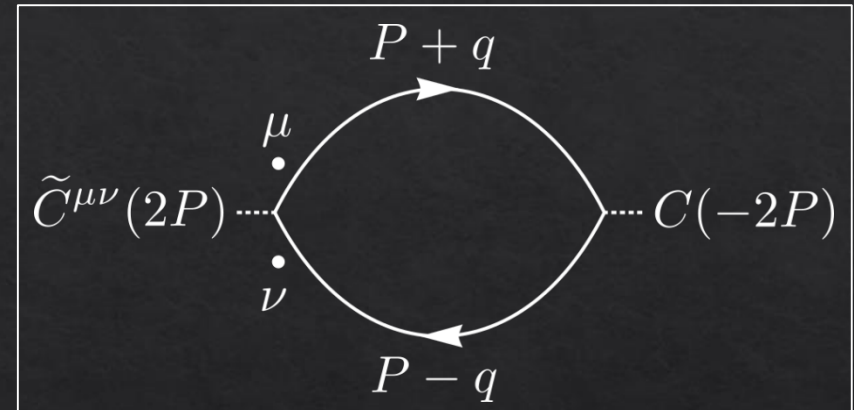
$1/N$ correction

◇ $M, M_{\mu\nu}, M_{\mu\nu\alpha\beta}$ correspond the following diagrams:

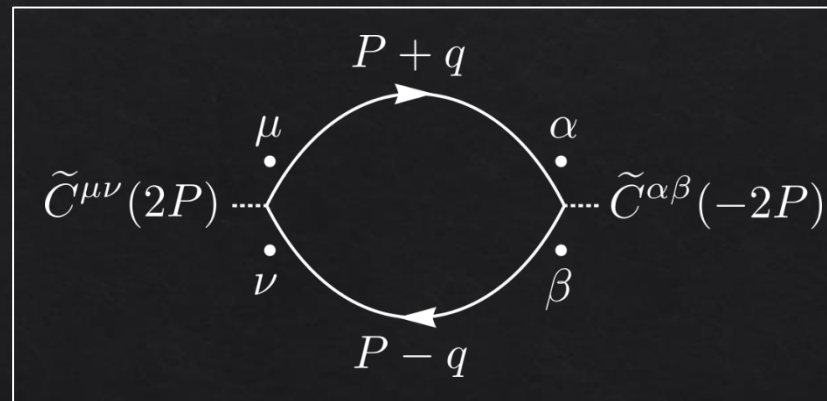
M



$M_{\mu\nu}$



$M_{\mu\nu\alpha\beta}$



Results

◆ After some calculation, we get

$$\tilde{C}^{\mu\nu} M_{\mu\nu\alpha\beta} \tilde{C}^{\alpha\beta} = -\frac{1}{12\pi} \left[p^2 + \left(\frac{3}{d-2} + \text{finite} \right) m_r^2 \right] (\tilde{C}^{\mu\nu})^2$$

$$-\frac{1}{12\pi p^4} \left[m_r^4 \frac{\tanh^{-1} \left(\frac{p}{\sqrt{m_r^2 + p^2}} \right)}{p \sqrt{m_r^2 + p^2}} + 2p^2 - 3m_r^2 \right] (p_\mu p_\nu \tilde{C}^{\mu\nu})^2$$

$$m_r^2 := (1 - 2\langle C \rangle) m^2$$

(with Dimensional Regularization: $d \rightarrow 2$)

◆ In the static frame ($p_1 = 0, p_2 = p$)

$$\tilde{C}^{\mu\nu} M_{\mu\nu\alpha\beta} \tilde{C}^{\alpha\beta} = -\frac{1}{12\pi} \left[p^2 + \left(\frac{3}{d-2} + \text{finite} \right) m_r^2 \right] B^2 + \dots$$

(Note: no other terms in S_{eff} include B)

$$\tilde{C}^{\mu\nu} =: \frac{1}{\sqrt{2}} \begin{pmatrix} A & B \\ B & -A \end{pmatrix}$$

Comments on Results

$$\frac{\mathcal{L}}{N} = \frac{1}{2}(\partial_\mu \vec{\phi})^2 + \frac{m^2}{2}\vec{\phi}^2 - m^2 C \vec{\phi}^2 + \tilde{C}^{\mu\nu} \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi} - \frac{1}{\alpha} C^2 + \frac{1}{2\alpha} (\tilde{C}^{\mu\nu})^2$$

◇ Comments

$$\tilde{C}^{\mu\nu} M_{\mu\nu\alpha\beta} \tilde{C}^{\alpha\beta} = -\frac{1}{12\pi} \left[p^2 + \left(\frac{3}{d-2} + \text{finite} \right) m_r^2 \right] B^2 + \dots$$

1. **Logarithmic** divergence (the coefficient of p^2) does **not appear**.

➤ This can be understood from Liouville action.

The rewritten Lagrangian is similar to Polyakov action with

$$g^{\mu\nu} \sim \delta^{\mu\nu} + \tilde{C}^{\mu\nu}$$

2. B (Vector particle) **violates the positivity** of state space.

(the coefficient of p^2 is **negative**.)

➤ If finite renormalization terms are added to recover the positivity, these should be non-local form like:

$$\mathcal{L}_{\text{c.t.}} \ni \alpha^2 T^{\mu\nu} \frac{\partial^2}{1 + \alpha \delta_Z \partial^2} T_{\mu\nu} \leftrightarrow \delta_Z \tilde{C}^{\mu\nu} \partial^2 \tilde{C}_{\mu\nu}$$

Summary and Discussion

- ◊ We studied the deformation of $O(N)$ vector model in the large N limit, with $1/N$ expansion.

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 + \frac{m^2}{2} \vec{\phi}^2 + \alpha \det(T_{\mu\nu})$$

- ◊ The deformation parameter should be **renormalized**.
- ◊ **No logarithmic divergence** appeared.
- ◊ Positivity of state space and locality of counterterms are **incompatible**.

Summary and Discussion

◆ Future direction

1. Considering **higher order** $\left(\frac{1}{N}\right)^n$ **corrections**.

Can these save the violation of the positivity?

2. The **full solution** to the $T\bar{T}$ flow equation for free $O(N)$ vector Lagrangian is known. [Bonelli-Doroud-Zhu, Cavaglià-Negro-Szécsényi-Tateo]

$$\mathcal{L}^{(\alpha)} = \frac{1}{2\alpha} \left[-1 + \sqrt{\left(1 + 2\alpha(\partial\vec{\phi} \cdot \bar{\partial}\vec{\phi})\right)^2 - 4\alpha^2(\partial\vec{\phi})^2(\bar{\partial}\vec{\phi})^2} \right]$$

(This takes the form of the Nambu-Goto action in $D=N+2$ target space dimensions in the static gauge.)

Does higher order terms of α change the results?

Thank you!

Plan of talk

◇ Introduction

◇ $T\bar{T}$ deformation in large N limit

- $N \rightarrow \infty$ limit
- $\frac{1}{N}$ correction

◇ Summary and Discussion

Back up

◇ Calculation of the effective action

$$\begin{aligned} \frac{S_{\text{eff}}}{N} = & -\text{tr} \left[(-\partial^2 + (1 - 2\langle C \rangle)m^2)^{-1} (m^2 \tilde{C} + \partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) \right]^2 \\ & + \int_x \frac{-1}{\alpha} \tilde{C}^2 + \frac{1}{2\alpha} (\tilde{C}^{\mu\nu})^2 \end{aligned}$$